B.TECH CSE | III YEAR MINI-PROJECT

SignVaani

Real-Time Indian Sign Language to Speech

1-Week Execution Blueprint

Day-by-day plan, design diagrams, test plan, report & viva kit

Team: _____

Roll numbers: _____

Guide: _____ Section: _____

Review dates: Week start _____ Final review _____

Version 1.0 | October 2026

Contents

- 1. How to use this blueprint
- 2. Frozen scope (what is and is not built)
- 3. Team roles
- 4. Software requirements specification (SRS)
- 5. System design: architecture, DFD, UML
- 6. Environment setup (Day 0 / Day 1 morning)
- 7. Day-by-day plan with exit criteria (Days 1-7)
- 8. Test plan and test cases
- 9. Results templates (fill with your numbers)
- 10. Risk register and fallback plan
- 11. Report chapter outline
- 12. Presentation (PPT) outline
- 13. 3-minute live demo script
- 14. Viva questions with model answers
- 15. Final submission checklist
- Appendix A: Recording protocol and consent form
- Appendix B: Daily stand-up log template

1. How to use this blueprint

This plan turns the SignVaani idea into a working, evaluated, documented mini-project in **7 days** (about 8 focused hours per day for a team of 4). The scope is deliberately cut down from the 8-week version so that every promised feature can actually be finished, tested and defended in the viva.

- Every day ends with **exit criteria**: measurable conditions that must be true before moving on. If a criterion fails, use the matching fallback in Section 10 instead of carrying the problem forward.
- Fill every **[XX]** with values you measured yourself. Never copy example numbers into the report.
- Keep a daily log (Appendix B). Examiners often ask "what went wrong and how did you fix it?"; the log answers that.
- Commit to GitHub at the end of every day with a clear message, so the history shows steady progress.

Golden rule for strict evaluation: it is better to present 15 signs that work reliably on a person the model has never seen, with honest numbers, than 50 signs that only work on the person who recorded them.

2. Frozen Scope

Freeze this on Day 1 morning and do not add features until Day 7 is complete.

2.1 Vocabulary: 15 hospital signs + IDLE (16 classes)

Group	Signs	Why chosen
People	ME, DOCTOR	Subject and the most-needed person
Symptoms	PAIN, FEVER	Most common hospital complaints
Body parts	HEAD, STOMACH	Location-based signs (tests body-relative features)

Group	Signs	Why chosen
Needs	WATER, MEDICINE, TOILET, HELP, NEED	Basic requests
Actions / questions	CALL, WHERE	Allow requests and questions
Answers	YES, NO	Needed in every conversation
Non-sign	IDLE	Resting hands, scratching, adjusting hair: prevents false triggers

Learn each sign from the **ISLRTC ISL dictionary** videos and, if at all possible, have an ISL user or teacher of the deaf confirm them on Day 1. Note the exact variant you used in the report.

2.2 Sentences the demo must produce

#	Signed (ISL gloss order)	English output	Hindi output (app shows Devanagari)
S1	ME HEAD PAIN	My head hurts.	Mere sir mein dard hai.
S2	ME STOMACH PAIN	My stomach hurts.	Mere pet mein dard hai.
S3	ME FEVER	I have a fever.	Mujhe bukhaar hai.
S4	ME NEED WATER	I need water.	Mujhe paani chahiye.
S5	ME NEED MEDICINE	I need medicine.	Mujhe dawa chahiye.
S6	ME NEED TOILET	I need to use the toilet.	Mujhe shauchaalay jaana hai.
S7	TOILET WHERE	Where is the toilet?	Shauchaalay kahaan hai?
S8	DOCTOR CALL	Please call the doctor.	Kripya doctor ko bulaiye.
S9	HELP	Please help me.	Kripya meri madad kijiye.
S10	YES / NO	Yes. / No.	Haan. / Nahin.

Hindi is written here in Latin letters for readability; the app displays and speaks it in Devanagari (see signvaani/rules.json). Get every Hindi sentence checked by a native speaker.

2.3 In scope vs out of scope

In scope (must be finished)	Out of scope (say so clearly in report)
Isolated recognition of 16 classes from a laptop webcam	Continuous / open-vocabulary ISL translation
Automatic sign start/end detection (no button per sign)	Fingerspelling A-Z
Rule-based ISL-order to English sentence + Hindi template	Neural translation, LLM sentence polishing
Speech: English offline (pyttsx3), Hindi via gTTS	Fully offline Hindi TTS (future scope: Indic-TTS)
Signer-independent evaluation (leave-one-signer-out)	Large user study / clinical validation
Own dataset: 5 signers x 16 classes x 10 takes = 800 clips	Mobile app

2.4 Measurable targets

Metric	Target	How measured
Top-1 accuracy, signer-independent (LOSO mean)	≥ 80%	train --loso, 5 folds
Top-1 accuracy, signer-dependent (random split)	≥ 95%	train --random-split
Live sign accuracy on an unseen person	≥ 70% of 60 attempts	Test case TC-12
False triggers while idle	≤ 1 per minute	Test case TC-10
Processing speed	≥ 15 FPS on CPU	FPS shown in app
Sign end to text on screen	≤ 500 ms	Test case TC-14
Sentence correctness for S1-S10	10 / 10	Test cases TC-07, TC-08

3. Team Roles

Member	Role	Owns	Also supports
A	Data lead	Vocabulary, recording sessions, consent forms, data QA, datasheet	Testing on Day 6
B	ML lead	Training, LOSO, model comparison, ablation, ONNX export, results tables	Data QA
C	Real-time & NLP lead	Spotter tuning, sentence rules, Hindi templates, TTS, live app	Demo video
D	Docs & QA lead	SRS, diagrams, test cases, report, PPT, README, daily log	Recording volunteers

Team of 2-3? Merge A+D and B+C. **Everyone** must be able to explain every module in the viva; spend 30 minutes each evening walking each other through the day's work.

4. Software Requirements Specification (SRS)

4.1 Functional requirements

ID	Requirement	Verified by
FR-1	Capture live video from a webcam at ≥ 15 FPS.	TC-01
FR-2	Extract pose, both-hand and face landmarks for every frame.	TC-02
FR-3	Fill short tracking gaps and normalise landmarks for position and body size.	TC-03, unit tests
FR-4	Detect the start and end of a sign automatically from hand motion.	TC-09
FR-5	Classify each detected sign into one of 16 classes with a confidence score.	TC-05
FR-6	Reject the IDLE class and predictions below the confidence threshold (0.8).	TC-10, TC-11
FR-7	Show the skeleton, top-3 predictions, recognised signs and sentence on screen.	TC-13
FR-8	Convert ISL gloss sequences into grammatical English and Hindi sentences.	TC-07, TC-08
FR-9	Speak the sentence on demand (SPACE) or automatically after a pause.	TC-15
FR-10	Allow undo, clear and language switch from the keyboard.	TC-16
FR-11	Provide a guided recorder that saves keypoints only and rejects poor-quality takes.	TC-04
FR-12	Train and evaluate models with signer-independent (LOSO) splits.	TC-06

4.2 Non-functional requirements

ID	Category	Requirement
NFR-1	Performance	≥ 15 FPS and ≤ 500 ms from sign end to text on an Intel i5 / Ryzen 5 laptop CPU, no GPU.
NFR-2	Offline	Recognition and English speech work without internet.
NFR-3	Privacy	No video is saved by the app or recorder; only landmark coordinates (.npy).
NFR-4	Usability	Large text overlay; single-key controls; status indicator (ready / signing).
NFR-5	Robustness	Works for signers not in the training set and for normal indoor lighting.
NFR-6	Maintainability	Modular Python package with unit tests; vocabulary and sentence rules in JSON.
NFR-7	Portability	Runs on Windows 10/11, Ubuntu 22.04 and macOS with Python 3.10 or 3.11.

4.3 Hardware and software requirements

Item	Minimum	Used by us (fill in)
CPU	Intel i5 8th gen / Ryzen 5	[XX]
RAM	8 GB	[XX]
Camera	720p webcam, front-facing	[XX]
OS	Windows 10/11, Ubuntu 22.04, macOS 12+	[XX]
Python	3.10 or 3.11 (MediaPipe 0.10.14 wheels)	[XX]
Libraries	mediapipe, opencv-python, numpy, torch (CPU), onnxruntime, scikit-learn, matplotlib, pyttsx3, gTTS, pygame	-
Training	Laptop CPU is enough for 800 clips (about 2-5 min per model); Google Colab optional	[XX]

5. System Design

Draw these diagrams again in draw.io or StarUML for the report (examiners expect editable, neat diagrams). The versions below are the exact content to reproduce.

5.1 Architecture

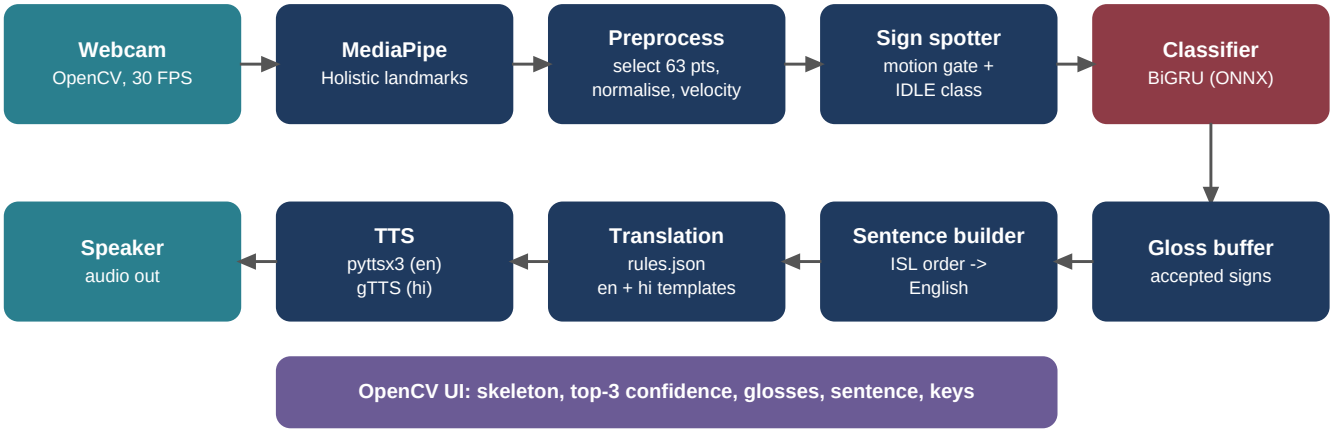


Figure: System architecture. Top row runs every frame; bottom row runs once per accepted sign or sentence.

5.2 Module description

Module (file)	Input	Output	Key logic
landmarks.py	MediaPipe result	(63, 3) array per frame	Keep 9 pose + 12 face + 2x21 hand points; NaN if missing
landmarks.preprocess	(T, 63, 3) clip	(32, 378) features	Interpolate gaps ≤ 6 frames, resample to 32, centre on shoulders, divide by shoulder width, add velocity
augment.py	raw clip	perturbed clip	Crop, speed 0.8-1.2x, rotate ±15°, scale, shift, noise, hand drop-out
models.py	(B, 32, 378)	class logits	BiGRU + attention (main); Transformer and 1D-CNN for comparison
spotter.py	frame stream	(sign, confidence)	Wrist-speed energy opens/closes window; reject IDLE and p < 0.8; cool-down
sentence.py + rules.json	gloss list	en / hi sentence	Category patterns (e.g. PRONOUN BODY PAIN); fallback word-by-word
speech.py	text, language	audio	pyttsx3 offline (en), gTTS + cache (hi), background thread
record.py	webcam	.npy clips	Countdown, 2.5 s capture, quality checks, retry
train.py / evaluate.py	dataset	model, metrics	LOSO / random split, early stopping, confusion matrix

5.3 Feature vector (per frame)

63 landmarks × (x, y, z) = 189 position values, plus 189 velocity values (difference from previous frame) = **378 features per frame**. Each sign window is resampled to **32 frames**, so the model input is 32 × 378. Normalisation:

```
p' = (p - shoulder_centre) / shoulder_width    v_t = p'_t - p'_(t-1)
```

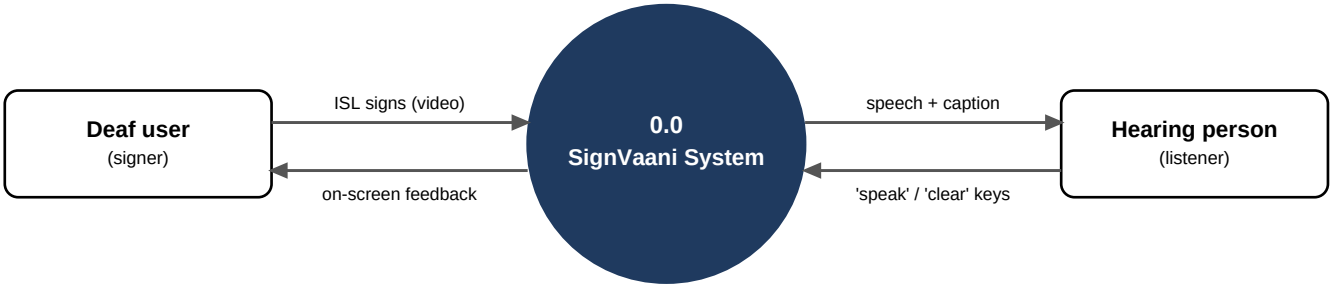


Figure 2: Context diagram (DFD Level 0)

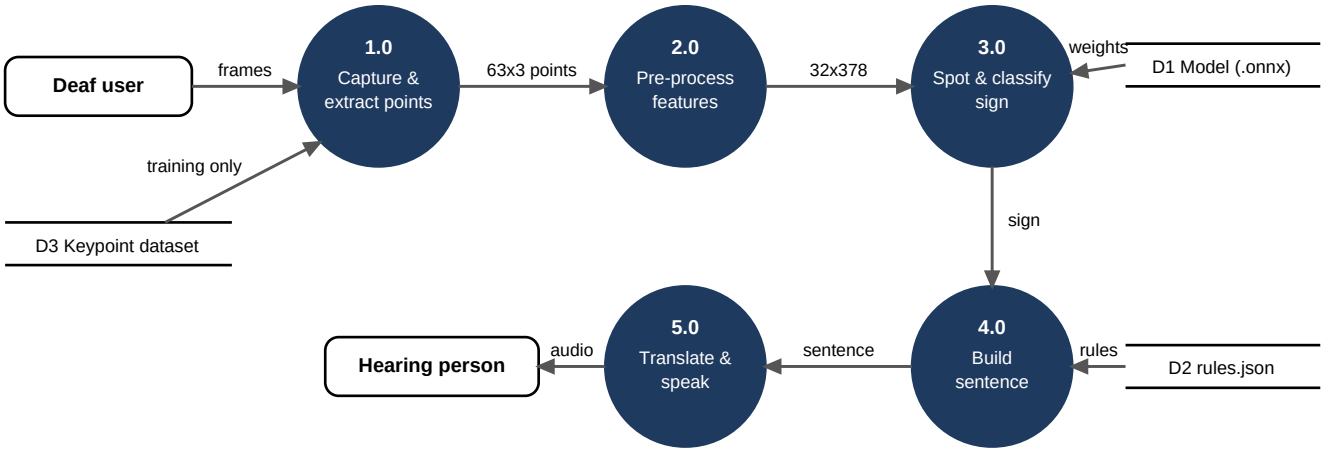


Figure 3: DFD Level 1

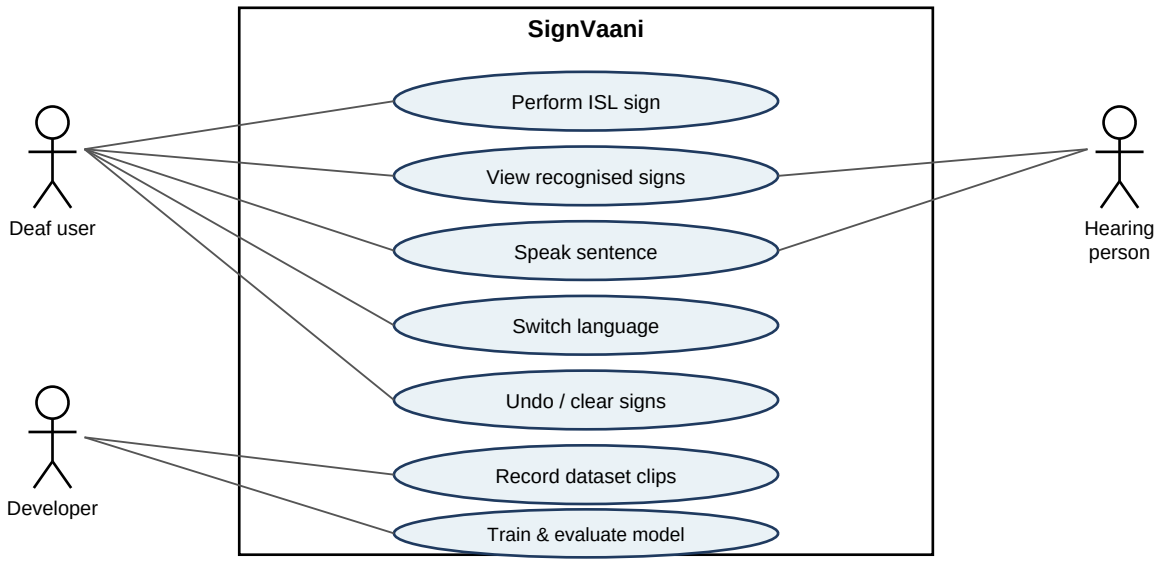


Figure 4: Use-case diagram

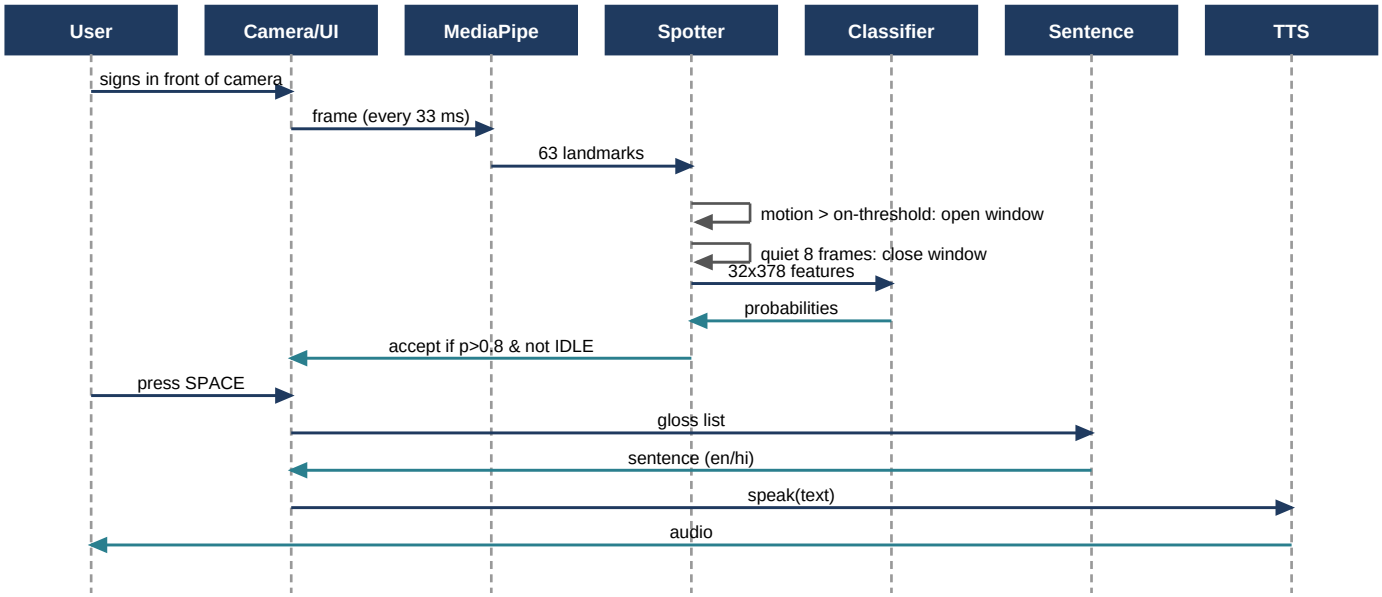


Figure 5: Sequence diagram (one sign, then speak)

5.4 Model architecture (main model)

Layer	Output shape	Parameters (approx.)
Input	32 x 378	0
Linear 378->256 + LayerNorm + GELU + Dropout 0.2	32 x 256	97 K
BiGRU, 2 layers, hidden 128, dropout 0.3	32 x 256	395 K
Attention pooling (Linear 256->1, softmax over time)	256	0.3 K
Dropout 0.3 + Linear 256->16	16	4 K
Total		about 0.5 M (verify: sum(p.numel() for p in model.parameters()))

Training: AdamW (lr 1e-3, weight decay 0.01), cosine schedule, cross-entropy with label smoothing 0.1, batch 64, up to 120 epochs with early stopping (patience 25) on validation accuracy.

6. Environment Setup (Day 1, first 2 hours)

Every team member installs the same versions. Use Python **3.10 or 3.11**; MediaPipe does not support every newer Python release.

```
# 1. Create and activate a virtual environment
python -m venv .venv
.venv\Scripts\activate            # Windows
source .venv/bin/activate          # Linux / macOS

# 2. Install pinned dependencies
pip install mediapipe==0.10.14 opencv-python==4.10.0.84 "numpy<2" scikit-learn matplotlib
pip install torch --index-url https://download.pytorch.org/whl/cpu
pip install onnx onnxruntime pyttsx3 gTTS pygame pytest

# 3. Check the install
python -c "import mediapipe, cv2, torch, onnxruntime; print('OK', mediapipe.__version__, torch.__version__)"
```

Project structure

```
ISL/
|-- configs/vocabulary.json      # frozen 16-class list
|-- signvaani/
|   |-- landmarks.py   augment.py   data.py   models.py
|   |-- train.py   evaluate.py   export_onnx.py   inference.py
|   |-- spotter.py   sentence.py   rules.json   speech.py
|   |-- record.py   app.py   make_synthetic.py
|-- tests/                       # pytest unit + pipeline tests
|-- data/keypoints/<SIGN>/<signer>_<take>.npy   # NOT committed if consent forbids
|-- models/                      # best.pt, sign_model.onnx, sign_model.json
|-- runs/                        # LOSO json, confusion matrices
|-- docs/                        # report, PPT, diagrams, datasheet
|-- requirements.txt   README.md   .gitignore
```

Core commands used during the week

```
python -m signvaani.record --signer s01 --takes 10             # record (Day 1-2)
python -m signvaani.train --data data/keypoints --loso         # main result (Day 3-4)
python -m signvaani.train --data data/keypoints --random-split  # signer-dependent result
python -m signvaani.train --data data/keypoints --test-signers s05 --val-signers s04
python -m signvaani.evaluate --ckpt models/best.pt --signers s05
python -m signvaani.export_onnx --ckpt models/best.pt          # Day 4
python -m signvaani.app --model models/sign_model.onnx --lang en  # Day 5+
python -m pytest -q                                             # every day before commit
```

7. Day-by-Day Plan

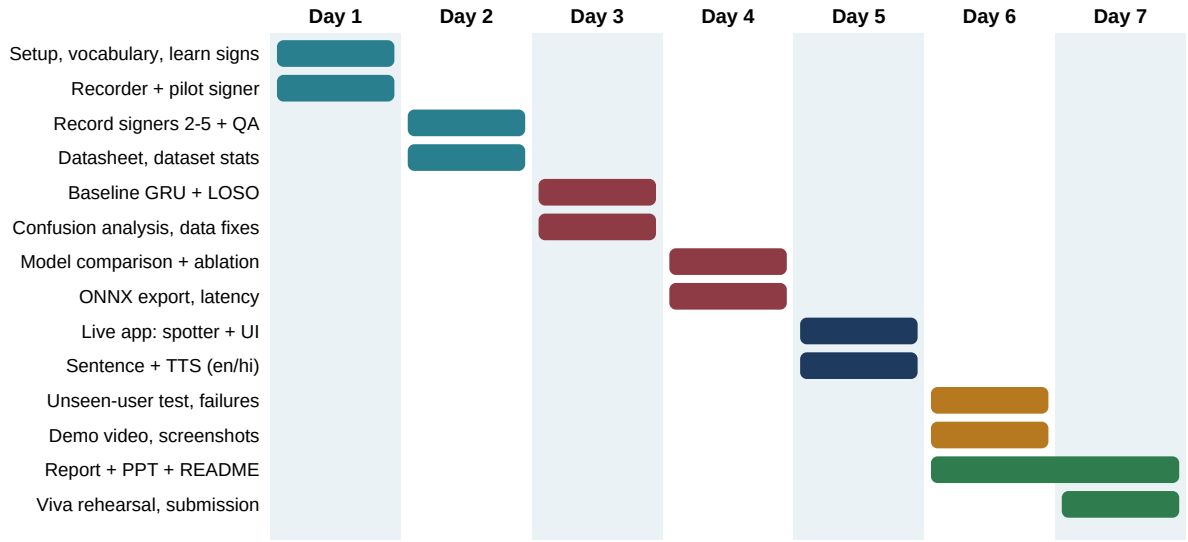


Figure 1: 7-day schedule (Gantt chart)

Times are a guide for an 8-hour day. Lines with the same time slot run in parallel. Do not start the next day until the exit criteria are met or the matching fallback (Section 10) has been applied.

Day 1 - Setup, vocabulary, recorder, pilot data

Time	Who	Task
09:00-11:00	All	Environment setup (Section 6); create GitHub repo; push skeleton.
09:00-11:00	A, D	Freeze vocabulary; download ISLRTC videos for the 15 signs; practise each sign.
11:00-13:00	B, C	Run the recorder; verify landmarks are saved with NaN for missing hands; check file layout.
11:00-13:00	D	Write SRS (Section 4) and draw architecture + DFD 0 in draw.io.
14:00-16:00	A	Contact ISL user / special school teacher to verify signs (phone or video call is fine).
14:00-17:00	All	Pilot recording: signer s01, 16 classes x 10 takes (about 30 min); fix lighting/camera position.
17:00-18:00	B	Inspect pilot data: plot wrist trajectories for 3 signs; confirm hands detected ≥ 90% of frames.

Exit criteria

- Every member's machine prints OK in the install check.
- data/keypoints has 16 folders x 10 clips for s01; recorder rejected bad takes.
- Vocabulary and sentence list frozen and committed.

Day 2 - Full dataset and quality control

Time	Who	Task
09:00-13:00	A, D	Record signers s02-s05 (about 40 min each incl. breaks). Vary clothing, background, lighting, distance (1-2 m).
09:00-11:00	B	Write a dataset-stats script: clips per class/signer, mean length, % frames with hands.
11:00-13:00	C	Implement/verify sentence rules for S1-S10 in rules.json; write unit tests.
14:00-16:00	A, B	Quality control: remove clips with < 60% hand detection or wrong sign; re-record if a class has < 8 clips for a signer.
14:00-16:00	D	Use-case and sequence diagrams; start report Chapters 1-2.
16:00-18:00	A	Write DATASHEET.md: signers (anonymised), conditions, sign sources, consent, licence.

Exit criteria

- About 800 clips: 5 signers x 16 classes x ≥ 8 good takes each.
- Dataset stats table and datasheet committed.
- Sentence unit tests pass for S1-S10.

Day 3 - Baseline model and honest evaluation

Time	Who	Task
09:00-10:00	B	Smoke-test pipeline on synthetic data (make_synthetic) to rule out code bugs.
10:00-13:00	B	Train BiGRU: random split, then LOSO (5 folds). Save runs/*.json.
10:00-13:00	C	Run the live app with the first checkpoint; tune spotter thresholds by watching the motion value.
14:00-16:00	B, A	Confusion matrix on each LOSO fold; list top-5 confused pairs; inspect those clips.
16:00-18:00	A	Fix data causes (wrong label, inconsistent sign execution); re-record specific takes if needed.
14:00-18:00	D	Report Chapter 3 (SRS) and Chapter 4 (design) with diagrams.

Exit criteria

- Baseline LOSO mean accuracy recorded (any value) with per-signer numbers.
- Confused pairs identified with a reason for each.
- If LOSO < 60%: apply fallback F2/F3 before Day 4.

Day 4 - Improve, compare, ablate, export

Time	Who	Task
09:00-12:00	B	Compare models with LOSO: GRU vs Transformer vs 1D-CNN (same data, same seed).
12:00-14:00	B	Ablation with the best model: hands only / +pose / +face / no augmentation (Section 9 table).
14:00-15:00	B	Train final model on all signers except a held-out validation signer; export ONNX; verify ONNX output = PyTorch output.
09:00-15:00	C	Live app: status indicator, top-3 bars, undo/clear/language keys, auto-speak after pause.
15:00-18:00	C, B	Measure FPS and latency (TC-01, TC-14) on the weakest laptop in the team.
09:00-18:00	D	Write test cases document (Section 8); Chapter 5 (implementation) draft.

Exit criteria

- Model comparison and ablation tables filled.
- models/sign_model.onnx + .json committed (or stored if large); app runs from ONNX.
- FPS ≥ 15 on the weakest laptop, otherwise use fallback F5.

Day 5 - Integration: sentence and speech

Time	Who	Task
09:00-12:00	C	Integrate SentenceBuilder + speech; check S1-S10 end to end in English and Hindi.
09:00-12:00	A	Record 2-3 more takes per class from a sixth person (s06) ONLY for final testing, not training.
12:00-14:00	B	Run evaluate.py on s06 offline clips; record the result.
14:00-17:00	All	Bug bash: each member uses the app for 20 minutes and logs every failure in the issue list.
17:00-18:00	C	Fix top bugs; freeze code by 18:00 (only bug fixes after this).

Exit criteria

- S1-S10 produce the correct English and Hindi sentence and speak aloud.

- Offline test-signer result recorded.
- Code freeze tag v1.0 on GitHub.

Day 6 - Testing with unseen users, results, demo video

Time	Who	Task
09:00-12:00	A, D	Execute all test cases TC-01 to TC-16; record pass/fail and evidence (screenshots).
09:00-12:00	C	Live test with s06 and if possible an ISL user: 60 sign attempts, 20 sentence attempts.
12:00-14:00	B	Final results tables and plots: LOSO bar chart, confusion matrix, latency table.
14:00-16:00	C, A	Record 3-minute demo video (Section 13) and a backup video for the viva.
14:00-18:00	D, B	Report Chapters 6 (testing) and 7 (results & discussion), limitations, future scope.

Exit criteria

- All test cases executed with evidence.
- Results tables contain only measured numbers.
- Demo video saved in two places (laptop + cloud drive).

Day 7 - Report, presentation, viva rehearsal, submission

Time	Who	Task
09:00-12:00	D, B	Finish report: abstract, conclusion, references (IEEE style), appendix; run plagiarism check.
09:00-12:00	C, A	PPT (Section 12); README with setup, usage, results, limitations, demo GIF.
13:00-15:00	All	Two full mock presentations with timing (10-12 min) + live demo on the presentation laptop.
15:00-17:00	All	Viva drill: each member answers 10 random questions from Section 14 without notes.
17:00-18:00	D	Final checklist (Section 15); submit; tag v1.1 on GitHub.

Exit criteria

- Report, PPT, code, dataset datasheet and demo video submitted.
- Every member can explain every module and every number.
- Live demo tested on the exact laptop, camera and room used for the review.

8. Test Plan and Test Cases

8.1 Test levels

Level	What	How
Unit	Normalisation, gap filling, resampling, augmentation, sentence rules, spotter logic	pytest (tests/); must pass before every commit
Integration	Record -> train -> export -> app pipeline	Synthetic-data pipeline test + real data
System	Live app against functional and non-functional requirements	Test cases below
Acceptance	Unseen person / ISL user performs S1-S10	TC-12, TC-15, user feedback form

8.2 System test cases

ID	Req.	Test	Expected result	Actual	P/F
TC-01	FR-1, NFR-1	Run app for 60 s, read FPS	≥ 15 FPS		
TC-02	FR-2	Stand 1.5 m, raise both hands	Skeleton on body and both hands		
TC-03	FR-3	Hide one hand for 0.1 s during a sign	Sign still recognised		
TC-04	FR-11	Record a take with hands out of frame	Recorder rejects and retries		
TC-05	FR-5	Perform each of 15 signs once	Correct label shown with confidence		
TC-06	FR-12	Run train --loso	Per-signer accuracy + mean/std saved		
TC-07	FR-8	Sign S1-S10 (English)	Exact sentences of Section 2.2		
TC-08	FR-8	Sign S1-S10 (Hindi)	Exact Hindi sentences of Section 2.2		
TC-09	FR-4	Sign 3 signs with short pauses	Exactly 3 signs detected		
TC-10	FR-6	Sit naturally, no signing, 3 min	≤ 3 false signs		
TC-11	FR-6	Perform a sign not in vocabulary	Rejected or low confidence		
TC-12	NFR-5	Unseen person, 60 sign attempts	≥ 70% correct		
TC-13	FR-7	Check overlay during use	Status, top-3, glosses, sentence visible		
TC-14	NFR-1	Film screen at 60 FPS; frames from hands-down to text	≤ 500 ms		
TC-15	FR-9	Press SPACE after S4; enable auto-speak	Sentence spoken in chosen language		
TC-16	FR-10	Press u, c, l keys	Undo, clear, language switch work		
TC-17	NFR-2	Disconnect internet, run English mode	Full pipeline works		
TC-18	NFR-3	Search project folders after a session	No video files created		

9. Results Templates (fill with measured values only)

Table R1: Dataset summary

Signers	Classes	Takes / class / signer	Total clips	Mean clip length	Frames with hands
[XX]	16	[XX]	[XX]	[XX] frames	[XX]%

Table R2: Signer-dependent vs signer-independent accuracy

Model	Random split top-1	LOSO top-1 (mean ± std)	LOSO macro-F1	Parameters	CPU ms / sign
BiGRU + attention	[XX]	[XX] ± [XX]	[XX]	[XX]	[XX]
Transformer	[XX]	[XX] ± [XX]	[XX]	[XX]	[XX]
1D-CNN	[XX]	[XX] ± [XX]	[XX]	[XX]	[XX]

Table R3: Per-signer LOSO accuracy (best model)

Test signer	s01	s02	s03	s04	s05	Mean
Top-1	[XX]	[XX]	[XX]	[XX]	[XX]	[XX]

Table R4: Ablation (LOSO, best model)

Configuration	Top-1	Change
Hands only	[XX]	-
Hands + pose	[XX]	[+XX]
Hands + pose + face	[XX]	[+XX]
Full model without augmentation	[XX]	[-XX]
Full model without velocity features	[XX]	[-XX]

To run the ablation, zero the excluded landmark groups in `preprocess` *(e.g. set the face slice to 0) behind a command-line flag, retrain with LOSO and record the mean.*

Table R5: Live system performance

FPS (weakest laptop)	Latency (TC-14)	False triggers / min	Unseen-person sign accuracy	Sentence success (S1-S10)
[XX]	[XX] ms	[XX]	[XX] / 60	[XX] / 10

Table R6: Most confused sign pairs

True sign	Predicted as	Count	Reason (hand shape / location / movement)	Fix tried
[XX]	[XX]	[XX]	[XX]	[XX]
[XX]	[XX]	[XX]	[XX]	[XX]

Expect LOSO accuracy to be lower than random-split accuracy. That gap is a finding, not a failure: explain it (different hand sizes, speeds, signing styles) and show what reduced it (normalisation, augmentation).

10. Risk Register and Fallback Plan

ID	Risk	Early warning	Fallback (apply immediately)
F1	MediaPipe install fails	Import error on Day 1	Use Python 3.10 exactly; pin mediapipe 0.10.14 and numpy<2; use a teammate's laptop for recording
F2	LOSO accuracy < 60% on Day 3	Confusion concentrated on a few pairs	Re-record confused signs consistently; replace one sign of each pair with a more distinct sign
F3	Too few clean clips	< 8 good takes per class per signer	Record 5 more takes for weak classes; add a 6th signer
F4	Many false triggers	TC-10 fails	Raise on-threshold and confidence; record 20 extra IDLE clips per signer

ID	Risk	Early warning	Fallback (apply immediately)
F5	Low FPS	< 15 FPS	MediaPipe model_complexity=0; reduce camera to 640x480; ONNX model
F6	Signs merge in live use	TC-09 fails	Increase quiet_frames; instruct users to drop hands briefly between signs
F7	Hindi TTS needs internet at review	No Wi-Fi in hall	Pre-generate audio for S1-S10 (cache) the day before; English works offline
F8	Team member unavailable	Missed stand-up	Roles overlap (Section 3); daily commits mean no work is lost
F9	Demo fails during review	Camera/lighting issue	Play the recorded demo video; then show unit tests and results live

11. Report Chapter Outline (40-60 pages)

Chapter	Contents	Pages
Front matter	Title page, certificate, declaration, acknowledgement, abstract (250 words), contents, list of figures/tables, abbreviations	8-10
1. Introduction	Background on ISL and the communication gap; problem statement; objectives; scope; report organisation	4-5
2. Literature Survey	Sensor vs vision vs pose-based SLR; ISL datasets (INCLUDE, CISLR, iSign); comparison table; research gap	6-8
3. Requirements (SRS)	Functional / non-functional requirements; hardware/software; feasibility (technical, economic, operational)	4-5
4. System Design	Architecture; DFD 0 and 1; use-case; sequence; class diagram of modules; model architecture; data format	6-8
5. Implementation	Dataset collection protocol; landmark extraction; normalisation; model and training; spotting; sentence rules; TTS; UI; key code snippets (short)	8-10
6. Testing	Test levels; unit tests; test case table with results; LOSO protocol	4-5
7. Results & Discussion	Tables R1-R6; confusion matrix; LOSO chart; screenshots; analysis of failures; limitations	6-8
8. Conclusion & Future Scope	What was achieved vs objectives; continuous translation, offline Indic TTS, mobile app, more signs	2
References	IEEE style; use the 24-paper literature survey PDF; cite only what you read	2
Appendices	Vocabulary with sign descriptions; consent form; datasheet; user manual; GitHub link	3-5

Strict-evaluation tips: number every figure and table and refer to each one in the text; write in the third person and past tense for results; keep code snippets short (full code goes on GitHub); paraphrase and cite all related work; keep similarity below your university's limit (usually 10-15%).

12. Presentation Outline (12 slides, 10-12 minutes)

#	Slide	Content	Speaker
1	Title	Project name, team, guide, department	D
2	Problem	Communication gap; ISL vs ASL tools; one real scenario (hospital)	A
3	Objectives & scope	4 objectives; what is in / out of scope	A
4	Literature & gap	4-5 key papers in one table; the gap SignVaani fills	D
5	Architecture	Architecture diagram; one line per module	C
6	Dataset	Vocabulary, signers, recording protocol, quality checks, datasheet	A
7	Features & model	63 landmarks, normalisation, 378-d vector; BiGRU + attention	B
8	Real-time spotting & sentences	Motion gate + IDLE class; ISL order -> English/Hindi	C
9	Results	Random vs LOSO accuracy; ablation; confusion pairs	B
10	Live demo	3-minute demo (Section 13)	C
11	Testing & limitations	Test case summary; honest limitations	D
12	Conclusion & future	Achievements vs objectives; future scope; GitHub link; thank you	D

13. 3-Minute Live Demo Script

Time	Action	What to say
0:00-0:20	Show app running, skeleton visible	"The app tracks 63 points on hands, body and face in real time on this laptop CPU, with no internet."

Time	Action	What to say
0:20-0:50	Sit still for 15 s	"While I am not signing, it stays at 'ready'. The IDLE class and motion gate prevent false words."
0:50-1:30	Sign S2 (ME STOMACH PAIN), press SPACE	"I signed in ISL order. The system detects each sign, shows confidence, and converts it to 'My stomach hurts.'"
1:30-2:00	Press I, sign S4 (ME NEED WATER), SPACE	"Now in Hindi: Mujhe paani chahiye."
2:00-2:30	Teammate (not in training data) signs S8	"This person was never in the training set. That is the signer-independent setting we evaluated."
2:30-3:00	Show undo and a deliberately wrong sign	"If a sign is wrong, the user can undo it. Our measured accuracy on unseen signers was [XX]%."

Rehearse in the review room the day before: check lighting from the front, a plain background, camera at chest height, 1.5 m distance, and the speaker volume. Keep the backup video open in another window.

14. Viva Questions with Model Answers

Every member should be able to answer all of these in their own words. Replace [XX] with your numbers.

Q1. Why did you use landmarks instead of raw video?

Landmarks reduce each frame from about 9 lakh pixel values to 189 coordinates, so a small model trains on a few hundred clips and runs in real time on a CPU. They are also unaffected by background, lighting and skin colour, and they protect privacy because no video is stored.

Q2. Why 63 landmarks and not all 543?

We kept the points ISL depends on: both hands (42), upper-body pose for sign location (9), and 12 face points around the lips, eyebrows and eyes for non-manual markers. The remaining face-mesh points add noise and computation without adding sign information. Our ablation table shows the effect of each group.

Q3. What is signer-independent evaluation?

The test signer never appears in training. We use leave-one-signer-out: with 5 signers we train 5 models, each tested on the one person left out, and report the mean and standard deviation. It reflects real use, where the user is always a new person.

Q4. Why is LOSO accuracy lower than random-split accuracy?

In a random split, clips from the same person appear in both training and test sets, so the model partly memorises that person's style. LOSO removes that advantage, so it is lower and more honest.

Q5. How do you normalise the landmarks?

Each frame is shifted so the midpoint of the shoulders is the origin and divided by the average shoulder width. This removes the effect of where the person stands and how big they appear. We also add velocity, the frame-to-frame difference, which captures motion.

Q6. How does the system know when a sign starts and ends?

It computes wrist speed relative to shoulder width every frame. When the smoothed speed goes above an on-threshold a window opens; when it stays below an off-threshold for 8 frames the window closes and is classified. An IDLE class and a 0.8 confidence threshold reject non-signs.

Q7. Why do you need an IDLE class?

Without it the classifier must pick one of the real signs for every movement, so scratching your face would be spoken as a word. IDLE gives the model a valid 'no sign' answer.

Q8. Why a GRU? Why not a Transformer?

With only about 800 clips, a GRU has a useful built-in bias for sequences and fewer parameters, so it overfits less. We trained a Transformer and a 1D-CNN with the same LOSO protocol and compared them in Table R2.

Q9. What is attention pooling?

Instead of using only the last GRU output, the model learns a weight for every frame and takes a weighted average, so it can focus on the frames where the hand shape is clearest.

Q10. How do you handle missing hands?

MediaPipe sometimes loses a hand during fast motion. Gaps of up to 6 frames are filled by linear interpolation; longer gaps are set to zero so the model knows the hand is absent. During training we randomly drop a hand for a few frames so the model learns to cope.

Q11. How is ISL grammar handled?

ISL often uses topic-first order without articles, e.g. ME STOMACH PAIN. We map glosses to categories (PRONOUN, BODY, PAIN) and match patterns to sentence templates, giving 'My stomach hurts.' Unmatched sequences fall back to a word-by-word sentence, which we show on screen.

Q12. What data augmentation did you use?

Random start/end crop, speed change 0.8-1.2x, rotation of plus or minus 15 degrees, scaling, shifting, small coordinate noise and hand drop-out. These simulate different signers, cameras and tracking errors.

Q13. What are the limitations?

Only 15 isolated signs from one domain, five or six signers, no full ISL grammar such as spatial referencing, regional sign variation not covered, and Hindi speech needs internet unless pre-cached. It is an aid, not a replacement for interpreters.

Q14. What is your contribution compared with existing work?

A complete, offline, CPU-only ISL-to-speech pipeline with automatic sign spotting, ISL-order sentence generation in English and Hindi, a small documented dataset, and signer-independent evaluation, which most student projects skip.

Q15. How did you ensure ethics and privacy?

Written consent from every signer, only anonymised keypoints stored, no video saved, signer IDs instead of names, and the signs verified with an ISL user.

Q16. What is the time complexity of inference?

Per frame, MediaPipe dominates. The classifier runs once per sign on a fixed 32 x 378 input, so its cost is constant per sign, a few milliseconds on CPU with ONNX Runtime.

Q17. Why ONNX?

ONNX Runtime is an optimised inference engine that is faster than plain PyTorch on CPU and does not need PyTorch installed in the final app. It also allows mobile deployment later.

Q18. How would you extend this to continuous sentences?

Train a sequence model with CTC loss or an encoder-decoder on a continuous ISL corpus such as ISLTranslate or iSign, so it outputs a gloss sequence without explicit sign boundaries.

15. Final Submission Checklist

Done	Item
[ ]	All unit tests pass (python -m pytest -q); screenshot in report appendix
[ ]	All test cases TC-01 to TC-18 executed with actual results and evidence
[ ]	Every number in report and PPT comes from runs/ files or test sheets (no placeholders left)
[ ]	LOSO per-signer and mean results reported next to random-split results
[ ]	Diagrams redrawn neatly, numbered and referenced in text
[ ]	References in IEEE format; every reference cited in text; plagiarism report within limit
[ ]	Consent forms signed and stored; datasheet complete; no faces or names in public data
[ ]	README: setup, usage, results, limitations, demo GIF/video link, team
[ ]	GitHub tagged v1.0 (code freeze) and v1.1 (final)
[ ]	Demo laptop: app tested in the review room; Hindi audio pre-cached; backup video ready
[ ]	Every member rehearsed the viva questions and can run the demo alone

Appendix A: Recording Protocol

- Camera at chest height, 1-2 m away, upper body and both hands fully in frame, light from the front.
- Each signer: 16 classes x 10 takes, 2.5 s per take, with a 3 s countdown. Start and end with hands resting.
- IDLE takes: hands resting, touching face, adjusting hair, looking at phone, small shrugs.
- Vary between signers: clothing colour, background, room, time of day, distance. Include left-handed signers if available.
- After each signer, run the dataset-stats script and re-record any class with fewer than 8 good takes.
- Never record anyone without signed consent. Use IDs s01, s02... never names in file names.

Consent form (template)

I, _____, agree to take part in the SignVaani mini-project at _____. I understand that the recording tool stores only body-landmark coordinates (no video or photographs), that my data will be identified only by a code (s__), and that it may be used for academic purposes and released publicly in anonymised keypoint form [ ] Yes [ ] No. I may withdraw at any time before submission by contacting the team.

Signature: _____ Date: _____ Signer ID: s___

Appendix B: Daily Stand-up Log Template

Date	Member	Done yesterday	Plan today	Blockers	Exit criteria met?
	A				
	B				
	C				
	D				